

MANOJ PARMAR

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Results-driven AI/ML Engineer with **6 years** building and shipping intelligent systems end to end. Has worked across the AI spectrum including **LLM-driven agentic workflows** (Claude, AWS Bedrock, ReAct tool-use), **recommender systems, and classical ML**. Deployed production agent and retrieval systems serving 30K+ users, with **responsible AI guardrails** and **eval-driven** release pipelines. Fluent in Python, SQL and AWS, with experience spanning **fintech and insurance**.

WORK EXPERIENCE

Prudential Insurance

Jersey City, NJ

Senior AI/ML Engineer

Feb 2024 – Present

- **Agentic AI Wellness Assistant**
 - Built a customer-facing **conversational AI assistant**, delivering personalized wellness guidance, content discovery and support routing for a healthcare insurance portal (**30K+ policyholders**), lifting **MAU by 20%**
 - Designed a **multi-stage agent harness** using **AWS Bedrock**, with **ReAct tool-use** across search, recommendation and escalation, plus short and long-term **conversational memory**; deployed as a containerized FastAPI service on AWS ECS Fargate
 - Built input/output **safety guardrails** - intent detection on user queries, post-response checks on model outputs, to block medical advice and auto-escalate to human care teams; shipped an **eval-driven CI pipeline** that scores every release for quality and safety pre-deployment
 - Reduced **inference cost by 40%** and **p95 latency by 30%** through model tiering (Haiku for safety checks, Sonnet for reasoning) and prompt caching
- **Personalized Recommendation and Ranking System**
 - Built a **personalization and ranking engine** that recommends health content and vendor services to 30K+ users, lifting content **CTR by 20%** and **MAU by 15%**
 - Designed the candidate generation and ranking pipeline: LLM-driven query expansion from user profiles, **hybrid semantic (embedding) and keyword retrieval** for candidate generation, and relevance ranking
 - Productionized the system as a scalable, event-driven pipeline on **AWS ECS** and **OpenSearch** with automated failure recovery, serving recommendations reliably at \$70 per 1K users
- Built an **experimentation framework** for wellness marketing campaigns, partnered with product and marketing to run 10+ **A/B tests** increasing enrollment 2x and projecting \$5M in annual claim savings

Anheuser-Busch InBev

Bangalore, India

Data Scientist, FinTech Products

Jul 2019 – Jul 2022

- Deployed dual **XGBoost credit risk models** in Azure Databricks scoring 1M invoices/month for ABInBev's B2B lending platform, driving **~\$9M** in annual benefits
- Designed an **experimentation framework (DiD)** for credit risk models in Brazil achieving 30% overdue reduction and 3% net revenue uplift; successfully launched a scale up and project was presented to CTO
- Built **daily PySpark pipelines** to ingest 1M+ invoices into Delta Lake tables; giving ML teams self-serve feature store access and cutting model build time 30%

TECHNICAL SKILLS

ML & AI: Agentic AI (ReAct, Tool Use, Planning), Vector DBs (FAISS, OpenSearch), LLMs, Recommender Systems, PyTorch, LangChain, LightGBM, XGBoost, Causal Inference, A/B Testing, Pandas, Scikit-learn

Languages & Tools: Python, SQL, AWS (Bedrock, ECS), Azure Databricks, FastAPI, Docker, Spark, PowerBI

LEADERSHIP & PROJECTS

Deputy Chair, AI/Data Analytics Group | Long-Term Care Insurance (LTCi Consortium)

Jun 2026 – Present

- Leading an industry-wide initiative across long-term care insurance carriers to advance AI capabilities and establish shared best practices for the sector

Food Image Recognition and Recipe Recommendation System | [Link](#)

Aug 2022 – Dec 2022

- **Classification** - Achieved **83%** accuracy on food recognition by training ~100K images using **CNN ResNet** models
- **Recommendation** - Built a recommendation layer to suggest similar food recipes using **collaborative filtering**

EDUCATION

MS, Computational Data Analytics, Georgia Institute of Technology (Atlanta, GA)

2022-2023

BS, Mechanical Engineering, Indian Institute of Technology, Bombay (India)

2015-2019